Case Study Title: Development of UniConnect – An Intelligent University Management Portal

1. Background

Nagarjuna University decides to develop a centralized web-based application called UniConnect to automate its academic and administrative activities. The university appoints a software development company to design and develop the application within a period of 12 months.

During the initial discussions, the university management provides a detailed set of requirements. The requirements are documented, reviewed, and approved by all stakeholders. The university also specifies a fixed budget, delivery schedule, and documentation standards.

Based on these conditions, the software development company selects the Waterfall Model for developing the application.

2. Initial Project Requirements

The proposed UniConnect application should provide the following functionalities:

Module	Functional Requirements
Student Management	Student registration, profile creation, admission details and academic history
Faculty Management	Faculty profiles, subject allocation and workload management
Attendance Management	Attendance entry, attendance calculation and reports
Examination Management	Examination schedules, marks entry and result publication
Fee Management	Fee details, payment records and receipt generation
Timetable Management	Class schedules, faculty timetables and classroom allocation
Administration	User management, role-based access and report generation

The university confirms that these requirements are sufficient for the initial release.

3. Changes in Project Requirements

After six months of development, the university management introduces several unexpected changes.

Change 1: Introduction of a Flexible Academic System

The university decides to implement a flexible credit-based academic system.

New requirements include:

- Students should be allowed to choose elective subjects.
- Students should be able to register for interdisciplinary courses.

- The system should automatically calculate credits earned.
- Students should be able to change their electives within a specified registration period.

Change 2: Integration with External Applications

The university decides to integrate UniConnect with external systems.

New requirements include:

- Integration with online payment gateways.
- Integration with biometric attendance devices.
- Integration with Moodle or other learning management systems.
- Integration with government scholarship portals.
- Integration with external placement platforms.

Change 3: New Stakeholder Expectations

During the demonstration of the initial application, students, parents, and faculty members provide additional feedback.

They request:

- A personalized student dashboard.
- A separate parent login to monitor attendance and academic performance.
- Automated notifications for low attendance.
- A grievance portal to submit and track complaints.
- Faculty announcements and assignment uploads.
- Real-time examination and result notifications.

Change 4: Changes in University Policies

The university introduces different attendance and evaluation policies across departments.

For example:

- Some departments require 85% attendance, while others require 75%.
- Different departments follow different internal assessment schemes.
- Students should be allowed to apply for attendance condonation.
- Examination grading rules may change based on academic regulations.

The development team realizes that the existing system design requires frequent modifications.

Change 5: Demand for Faster Delivery

The university management is no longer willing to wait until the completion of the entire application.

They demand that:

- Important features should be released as soon as they are ready.
- Students and faculty should be able to test new features.
- Feedback should be collected after every release.
- Critical changes should be implemented without redesigning the entire application.
- New features should be deployed frequently.

4. Current Situation

The development company now faces several challenges:

- The original requirements are no longer stable.
- The existing system design requires repeated modifications.
- New features are being requested even before the existing ones are completed.
- Stakeholders expect continuous interaction with the development team.
- The university wants working software delivered in smaller releases.
- The original development schedule is becoming difficult to maintain.

The project manager realizes that continuing with the existing development process may increase development time, cost, and the risk of delivering a system that does not meet the university's expectations.

The project manager calls a meeting with the development team to reconsider the software development approach.

5. Website Development Task

Based on the case study, design and develop a working prototype of the UniConnect web application.

Your prototype should include at least the following features:

- Student and faculty login.
- Student and faculty dashboards.
- Attendance management.
- Course and elective registration.
- Examination and result management.
- Fee management.
- Student grievance portal.

- Notifications and announcements.
- Parent dashboard.
- Administrator dashboard.